For instance, for Cauchy transitions on $\mathbb{X} = \mathbb{R} = \mathbb{Y}$ we have

$$\mathcal{Q}(x, dy) = \frac{1}{\pi a} \frac{1}{1 + ((x-y)/a)^2} \, dy \quad \text{with} \quad a > 0$$

$$\Longrightarrow W(x,y) = \log(a\pi) + \log\left(1 + ((x-y)/a)^2\right) \quad \text{and} \quad |\partial_x W(x,y)| \vee |\partial_y W(x,y)| \leqslant 1/a$$

$$\Longrightarrow c_w = \log(a\pi), \qquad \tau_{p,w} = 2/a^2 = \tau_{q,w} \quad \text{and} \quad p_w = q_w = 2.$$

5.2.1. *Exponential targets.* Consider some continuous target potential functions $U, V$ such that

$$U(x) \geqslant c_u + \tau_u \, \|x\|^{p_u} \quad \text{and} \quad V(x) \geqslant c_v + \tau_v \, \|x\|^{q_v} \quad \text{with } c_u, c_v \in \mathbb{R} \text{ and } p_u, q_v > 0.$$

Also assume that $(W_U, W^V)$ are continuous functions. Following the proof of Proposition 5.1, the exponential moment condition (5.1) ensures that (4.21) holds for any $0 < \delta < 1/2$ and the probability measures $\lambda_U$ and $\nu_V = \lambda_V$ have finite $p_w$ and $q_w$ absolute moments. By Jensen's inequality, we also have

$$\mathcal{U}_\delta(x) \quad \geqslant \quad \delta c_u + \delta \tau_u \, \|x\|^{p_u} - c_w - \log\left(1 + \tau_{p,w} \, \|x\|^{p_w} + \tau_{q,w} \, m_v^{q_w}\right) \stackrel{\|x\| \to \infty}{\longrightarrow} \infty$$

In the same vein we check that

$$\lim_{\|x\| \to \infty} (\delta U(x) - W^V(x)) = \infty = \lim_{\|x\| \to \infty} (\delta V(x) - W_U(x)). \tag{5.4}$$

We conclude that $\mathcal{H}'_\delta(U, V)$ is met for any $0 < \delta < 1/2$. Also note that $(U, V)$ satisfy (4.29) with the semi-metrics

$$\varrho_U(x_1, x_2) := \overline{\tau}_u \, \|x_1 - x_2\|^{p_u} \quad \text{and} \quad \varrho_V(x_1, x_2) := \overline{\tau}_v \, \|x_1 - x_2\|^{q_v}$$

for some parameters $\overline{\tau}_u, \overline{\tau}_v > 0$ so that the Wasserstein estimates (4.30) hold.

5.2.2. *Heavy tailed targets.* Consider continuous target potential functions $U, V$ such that

$$U(x) \geqslant c_u + \log\left(1 + \tau_u \, \|x\|^{p_u}\right) \quad \text{and} \quad V(x) \geqslant c_v + \log\left(1 + \tau_v \, \|x\|^{q_v}\right)$$

with $c_u, c_v \in \mathbb{R}$ and some parameters

$$p_u > 1 + 2p_w \quad \text{and} \quad q_v > 1 + 2q_w.$$

Condition (4.21) is met as soon as

$$(1 - 2\delta)p_u > 1 \quad \text{and} \quad (1 - 2\delta)q_v > 1$$

On the other hand, the probability measures $\lambda_U$ and $\lambda_V$ have finite $p_w$ and $q_w$ absolute moments as soon as

$$p_u - p_w > 1 \quad \text{and} \quad q_v - q_w > 1.$$

Finally, (5.4) holds when

$$\delta p_u - p_w > 0 \quad \text{and} \quad \delta q_v - q_w > 0.$$

This shows that $\mathcal{H}_\delta(U, V)$ for any $0 < \delta < 1/2$ such that

$$\frac{p_w}{p_u} < \delta < \frac{1}{2}\left(1 - \frac{1}{p_u}\right) \quad \text{and} \quad \frac{q_w}{q_v} < \delta < \frac{1}{2}\left(1 - \frac{1}{q_v}\right).$$

Also note that the exponential functions $(e^{\delta U}, e^{\delta V})$ satisfy (4.27) with the semi-metrics

$$\varrho_\varphi(x_1, x_2) := \overline{\tau}_u \, \|x_1 - x_2\|^{p_u} \quad \text{and} \quad \varrho_\psi(x_1, x_2) := \overline{\tau}_v \, \|x_1 - x_2\|^{q_v}$$

for some parameters $\overline{\tau}_u, \overline{\tau}_v > 0$ so that the Wasserstein estimates (4.28) hold.

5.3. **Some boundary states.** Consider a bounded Lipschitz domain $\mathbb{X} = \mathbb{Y} \subset \mathbb{R}^d$ equipped with the uniform measure $\lambda = \nu$. Denote by $\varrho_\partial(x, \partial\mathbb{X})$ the distance from $x \in \mathbb{X}$ to the boundary $\partial\mathbb{X}$. We assume that the target distribution $\lambda_U$ has a continuous potential function $U$ and satisfies the estimates

$$\lambda_U(dx) \leqslant c_{q,u}\ \varrho_\partial(x, \partial\mathbb{X})^{q_u}\ \lambda(dx)$$

for some parameters $c_{p,u}, c_{q,u} > 0$ and $q_u > 0$. Similarly, $\nu_V = \lambda_V$ has a continuous potential function $V$ and satisfies the above estimates for some $c_{p,v}, c_{q,v} > 0$ and $q_v > 0$. Finally, we assume that the integrated functions $(W^V, W_U)$ are continuous.

When $W(x, y)$ is bounded, Sinkhorn transitions are strongly mixing, in the sense that $\chi(\mathcal{S}) < 1$ with the uniform contraction parameter $\chi(\mathcal{S})$ defined in (4.2). The exponential decays of $\Phi$-entropies associated with these strong mixing models are discussed in Section 4.1. Note that for any $x \in \mathbb{X}$ we have

$$U(x) \geqslant -\log c_{q,u} + q_u\ \log \frac{1}{\varrho_\partial(x, \partial\mathbb{X})} \Longleftrightarrow e^{\delta U(x)} \geqslant \frac{1}{c_{q,u}^\delta}\ \frac{1}{\varrho_\partial(x, \partial\mathbb{X})^{\delta q_u}}.$$

By Remark 4.7 the above models satisfy $\mathcal{H}_\delta(U, V)$ for any $\delta \in ]0, 1[$. In addition, following the proof of (3.6) we check that the exponential function $e^{\delta U}$ satisfies (4.27) with the semi-metric

$$\varrho_\varphi(x_1, x_2) = \bar{c}_u\ \left(\frac{1}{\varrho_\partial(x_1, \partial\mathbb{X})} - \frac{1}{\varrho_\partial(x_2, \partial\mathbb{X})}\right)^{\delta q_u}\ 1_{x_1 \neq x_2} \quad \text{for some parameter } \bar{c}_u > 0.$$

Similarly, the exponential function $e^{\delta V}$ satisfies (4.27) with the metric $\varrho_\psi$ defined as $\varrho_\varphi$ by replacing $(c_u, q_u)$ by $(c_v, q_v)$, so that the Wasserstein estimates (4.28) hold.

When $W(x, y) = W(y, x)$ is symmetric and non necessarily bounded, assume we have an estimate of the form

$$W(x, y) \leqslant c_w + q_w\ \log \frac{1}{\varrho_\partial(x, \partial\mathbb{X})}$$

for some $c_w \in \mathbb{R}$ and some parameters $0 \leqslant q_w < (q_u \wedge q_v)/2$. In this case condition $\mathcal{H}_\delta(U, V)$ holds for any $\frac{q_w}{q_u \wedge q_v} < \delta < 1/2$.

5.4. **Beta marginal models.** Next we illustrate the boundary state models discussed in Section 5.3 with the open unit interval $\mathbb{X} = \mathbb{Y} = ]0, 1[$ with boundary $\partial\mathbb{X} = \{0, 1\}$. The distance to the boundary is given explicitly by the function

$$x \in \mathbb{X} \mapsto \varrho_\partial(x, \partial\mathbb{X}) = x\ 1_{]0,1/2[} + (1 - x)\ 1_{[1/2,1[}(x) \quad \text{and} \quad \lambda(dx) = 1_{]0,1[}(x)dx.$$

Note that

$$\frac{1}{2}\ \varrho_\partial(x, \partial\mathbb{X}) \leqslant x(1 - x) \leqslant \varrho_\partial(x, \partial\mathbb{X})$$

and consider the reference Markov transition $\mathcal{Q}$ defined by the semi-circle distribution (3.11). Let $(\lambda_U, \lambda_V)$ be some Beta marginal distributions given by

$$\lambda_U(dx) = \frac{1}{c_u}\ x^{a_u}(1 - x)^{b_u}\ \lambda(dx) \quad \text{and} \quad \lambda_V(dx) = \frac{1}{c_v}\ x^{a_v}(1 - x)^{b_v}\ \lambda(dx)$$

for some normalizing constants $c_u, c_v > 0$ and some parameters $a_u, a_v, b_u, b_v > 1$. In this context, we have

$$\frac{1}{2^{a_u \vee b_u} c_u}\ \varrho_\partial(x, \partial\mathbb{X})^{a_u \vee b_u}\ \lambda(dx) \leqslant \lambda_U(dx) \leqslant \frac{1}{c_u}\ \varrho_\partial(x, \partial\mathbb{X})^{a_u \wedge b_u}\ \lambda(dx).$$

Also observe that

$$\begin{aligned} 1 \leqslant \frac{1}{1 - (x - y)^2} &= \frac{1}{(1 - (x - y))(1 + (x - y))} \\ &= \frac{1}{y + (1 - x)}\ \frac{1}{x + (1 - y)} \leqslant \frac{1}{x(1 - x)} \wedge \frac{1}{y(1 - y)}, \end{aligned}$$

which yields the estimates

$$\begin{aligned} W(x,y) &= \log q(x) + \frac{1}{2}\log\left(\frac{1}{(1-(x-y))(1+(x-y))}\right) \\ &\leqslant \frac{1}{2}\log\left(\frac{1}{x(1-x)}\right) \quad \text{with} \quad \frac{1}{3} \leqslant q(x) := \int_0^1 \sqrt{1-(x-y)^2}dy \leqslant 1. \end{aligned}$$

Note that for any $(x,y) \in ]0,1[^2$ we obtain

$$|\partial_x \sqrt{1-(x-y)^2}| = \frac{|y-x|}{\sqrt{((1-x)+y)((1-y)+x)}} \leqslant g(y) := \frac{1}{\sqrt{y(1-y)}} \Longrightarrow \lambda(g) < \infty$$

as well as

$$|\partial_x \log(1-(x-y)^2)| \leqslant g(x)^2 \wedge g(y)^2.$$

In this example, the functions $(U,V)$ and $(W_U, W^V)$ are differentiable functions. Condition (4.19) is clearly met for any $0 < \delta < 1/2$ and we have

$$\delta U(x) - W^V(x) \geqslant \delta \log(c_u) + (\delta a_u - 1/2)\log\frac{1}{x} + (\delta b_u - 1/2)\log\frac{1}{1-x}.$$

This shows that $\mathcal{H}'_\delta(U,V)$ is met for any $((a_u \wedge b_u) \wedge (a_v \wedge b_v))^{-1} < 2\delta < 1$. We also have

$$U(x_1) + U(x_2) \geqslant \overline{c}_u : +\varrho_U(x_1,x_2)$$

with $\overline{c}_u := 2\log(c_u)$ and the metric

$$\varrho_U(x_1,x_2) := a_u \left|\log\frac{1}{x_1} - \log\frac{1}{x_2}\right| + b_u \left|\log\frac{1}{1-x_1} - \log\frac{1}{1-x_2}\right|.$$

We conclude that $U$ satisfies (4.29). Similarly $V$ satisfies (4.29) with a metric $\varrho_V$ defined as $\varrho_U$ by replacing $(a_u, b_u, c_u)$ by $(a_v, b_v, c_v)$.

Recall that $uv \geqslant (u+v)/2$ and $u^p v^q \geqslant (uv)^{p\wedge q}$ for any $u, v \geqslant 1$ and any parameters $p, q > 0$ we see that

$$\begin{aligned} e^{\delta U(x)} &\geqslant \frac{c_u^\delta}{2}\left(\left(\frac{1}{x}\right)^{\delta a_u} + \left(\frac{1}{1-x}\right)^{\delta b_u}\right), \\ e^{\delta U(x)} &\geqslant c_u^\delta \left(\frac{1}{x(1-x)}\right)^{\delta(a_u\wedge b_u)} \geqslant \overline{c}_u^\delta \frac{1}{\varrho_\partial(x,\partial\mathbb{X})^{\delta(a_u\wedge b_u)}} \quad \text{with} \quad \overline{c}_u := 2^{(a_u\wedge b_u)} c_u. \end{aligned}$$

Following the proof of (3.6) we check that the exponential function $e^{\delta U}$ satisfies (4.27) with

$$\varrho_\varphi(x_1,x_2) = \overline{c}_u \left(\frac{1}{\varrho_\partial(x_1,\partial\mathbb{X})} - \frac{1}{\varrho_\partial(x_2,\partial\mathbb{X})}\right)^{\delta(a_u\wedge b_u)}$$

as well as with the semi-metric

$$\varrho_\varphi(x_1,x_2) = \overline{c}_u \left(\left|\frac{1}{x_1} - \frac{1}{x_2}\right|^{\delta a_u} + \left|\frac{1}{1-x_1} - \frac{1}{1-x_2}\right|^{\delta b_u}\right)$$

for some parameter $\overline{c}_u > 0$. Similarly, the exponential function $e^{\delta V}$ satisfies (4.27) with the complete metric $\varrho_\psi$ defined as $\varrho_\varphi$ by replacing $(a_u, b_u, c_u)$ by $(a_v, b_v, c_v)$, so that the Wasserstein estimates (4.28) hold.